

Harsh Shah

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EDUCATION

Northeastern University , Boston, MA	Dec. 2025
Master of Science in Computer Science, GPA: 3.9/4.0	
Courses: Programming Design Paradigm, Foundations of Artificial Intelligence, Algorithms, Machine Learning, Database Management Systems, Machine Learning Operations, Natural Language Processing, Web Technology	
Savitribai Phule Pune University , Pune, India	June 2023
Bachelor of Engineering in Computer Engineering, GPA: 9.1/10.0	

SKILLS

Languages:	Python, Java, C++, C#, R
Web/Frameworks:	HTML, CSS, React, Node.js, .NET
Databases:	MySQL, Microsoft SQL Server, MongoDB, DynamoDB, Snowflake, Pinecone
ML & Data:	MLflow, TensorFlow, Keras, PyTorch, Scikit-Learn, NLTK, Apache Airflow, Grafana
Cloud/DevOps:	Git, GitHub, CI/CD, Microsoft Azure, Docker, Jenkins, Kubernetes, Prometheus, AWS S3 Buckets

WORK EXPERIENCE

Atlas SP Partners , New York City, New York	Feb. 2026 – Present
Associate Software Engineer	
- Collaborated in a cross-functional team to develop and implement full stack applications, automation workflows and centralized database storage for Residential Mortgage and Commercial Real Estate divisions of Securitized Products. - Architected and deployed a distributed AI document intelligence platform using Python, React, Docker, LangChain, and Azure Service Bus. Built a scalable RAG pipeline with vector embeddings and a vector database for semantic retrieval, enabling high-throughput processing of 100+ documents daily across multiple asset classes. - Developed a full-stack Commercial Real Estate (CRE) facility onboarding platform using React and Python, automating workflows for facilities, covenants, tranches and funding pools replacing manual Excel-based processes.	
Atlas SP Partners , New York City, New York	Jan. 2025 – Aug. 2025
Software Engineer Intern	
- Developed a full-stack daily margin call platform using .NET, React, Python, and RESTful APIs on Azure-hosted microservices, automating the Resi margin call workflow and reducing manual effort by 3–5 hours daily. - Built a config-driven, fully automated data ingestion pipeline in Python, integrating data from 3+ asset classes into Snowflake, enabling centralized, reliable storage and reducing Excel based system dependency by 70%. - Configured and managed automated workflow jobs in Control-M Helix to automate and orchestrate daily processes. - Developed a Python based automated solution for Resi Due Diligence, paired with an Excel interface via PyXll, reducing manual effort by 15-20 hours per deal. - Developed an email monitoring Python package using Microsoft Graph API to track mailboxes and send bulk emails.	
Knorr-Bremse Technology Center India Pvt Ltd , Pune, India	Apr. 2022 – Apr. 2023
Software Engineer Intern	
- Improved the efficiency and accuracy of the data management process by 70% by automating the process of MS SQL database management by developing a Python script. - Designed and maintained Microsoft SQL Server database schema to support various Software Engineering Key Performance Indicator (KPI) metrics following the Agile methodology for software development. - Configured and developed Grafana dashboards for data visualization of various KPI metrics by developing SQL queries.	

PROJECTS

Optimizing Bluebikes Operations with ML-Based Demand Prediction (link)	Oct. 2025 – Dec. 2025
- An end-to-end MLOps pipeline (Python, Docker, Airflow) for hourly bike-demand prediction using 40+ features, achieving $R^2 > 0.85$ with XGBoost, LightGBM and Random Forest models, tracked via MLFlow. - Automated production deployment with CI/CD, bias and drift monitoring (Evidently AI) with incremental data ingestion, reducing runtime by 70% with integrated release validation (Github-Actions) and alerting on Slack.	
Text Translation, Transformer and LSTM Architecture (link)	Jan. 2024 – Apr. 2024
Collaborated on a team project to develop a Python based French-to-English translation system using a Transformer model from scratch and achieving 91% translation accuracy to support global communication and cultural exchange.	
Image Manipulation and Forgery Detection, Image Segmentation Model and Deep Learning (link)	July 2022 – May 2023
Led a machine learning project on image manipulation detection model using the ResNet and Unet architecture achieving 93% accuracy on the CASIA dataset, while improving efficiency through a survey of existing methods.	